

Three-Model Comparison

Test-Card Performance Across Three Model Specifications

Overview

This report compares three residential AVM model runs on the `model.test_card` table. `meta_sale_price` is the truth column; `pred_card_initial_fmv` is the card-level prediction.

Run ID	Spec Label	Description
2026-02-11-recurring-rob	2026-02-11-recurring-rob	Final production model for modeling year 2026. Baseline for this comparison.
2026-03-31-gallant-jacob	logged_outcome_prod_prod_hyperparams	Same hyperparameters and architecture as <code>recurring-rob</code> ; only difference is sale price is logged during training.
2026-04-11-magical-manasi	cov_rho_3_hyperparam_search	Logged sale price, plus a new hyperparameter set found via CV search (selected per-model by best RMSE). Each individual model was trained with an added covariance penalty term.

All metrics and decile ratio curves are computed both overall (pooled across all Cook County triads) and separately by `meta_triad_code`.

Metric Table

The best value between the models listed in the chart is highlighted in green.

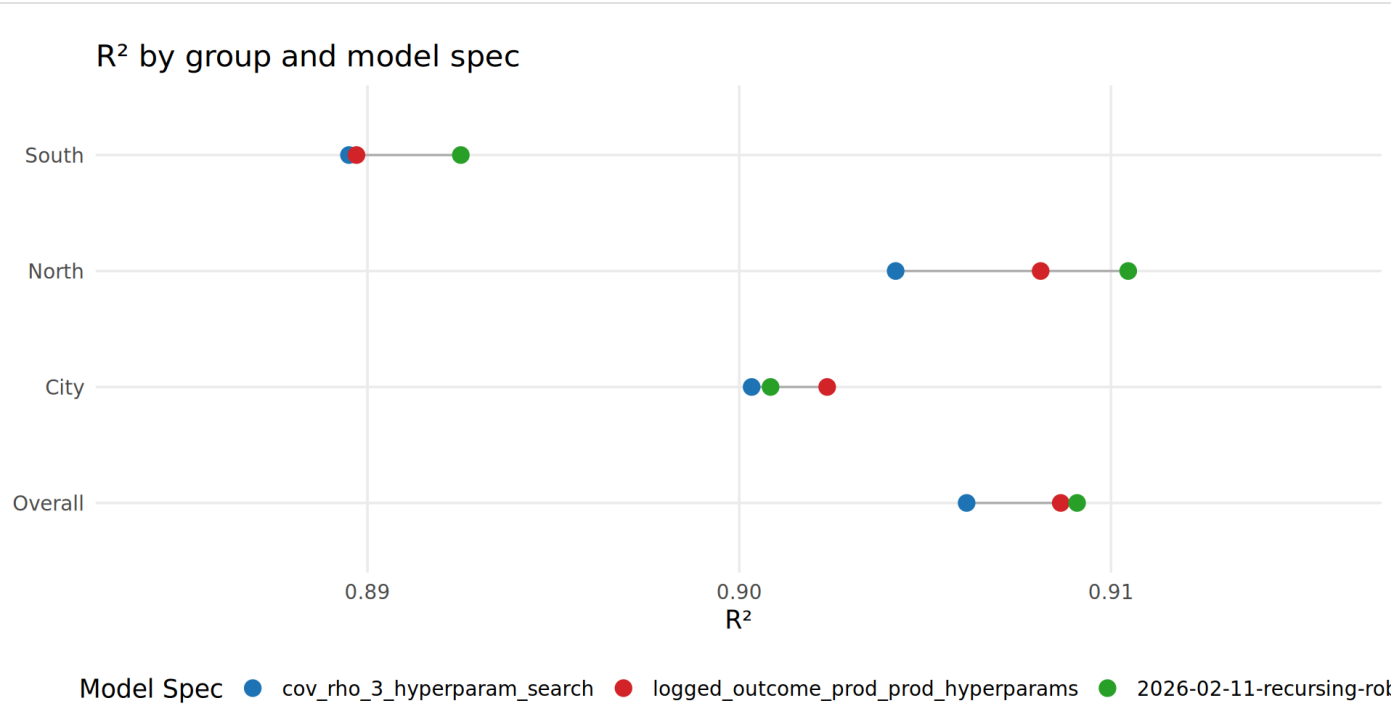
Overall	City	North	South
Metric	2026-02-11-recurring-rob	cov_rho_3_hyperparam_search	logged_outcome_prod_prod_hyperparams
N	36,524	36,524	36,524
R ²	0.909	0.906	0.909
RMSE	131,059	129,820	135,219
MAE	76,044	76,246	78,540

Metric	2026-02-11-recurring-rob	cov_rho_3_hyperparam_search	logged_outcome_prod_prod_hyperparams
MAPE %	21.54	20.695	20.675
MdAPE %	13.101	13.539	13.824
Med Ratio	0.966	0.967	0.944
COD	22.026	21.152	21.147
PRD	1.092	1.054	1.078
PRB	-0.135	-0.066	-0.101
MKI	0.891	0.949	0.904
VEI	-40.532	-17.92	-29.199

Per-Metric Comparisons

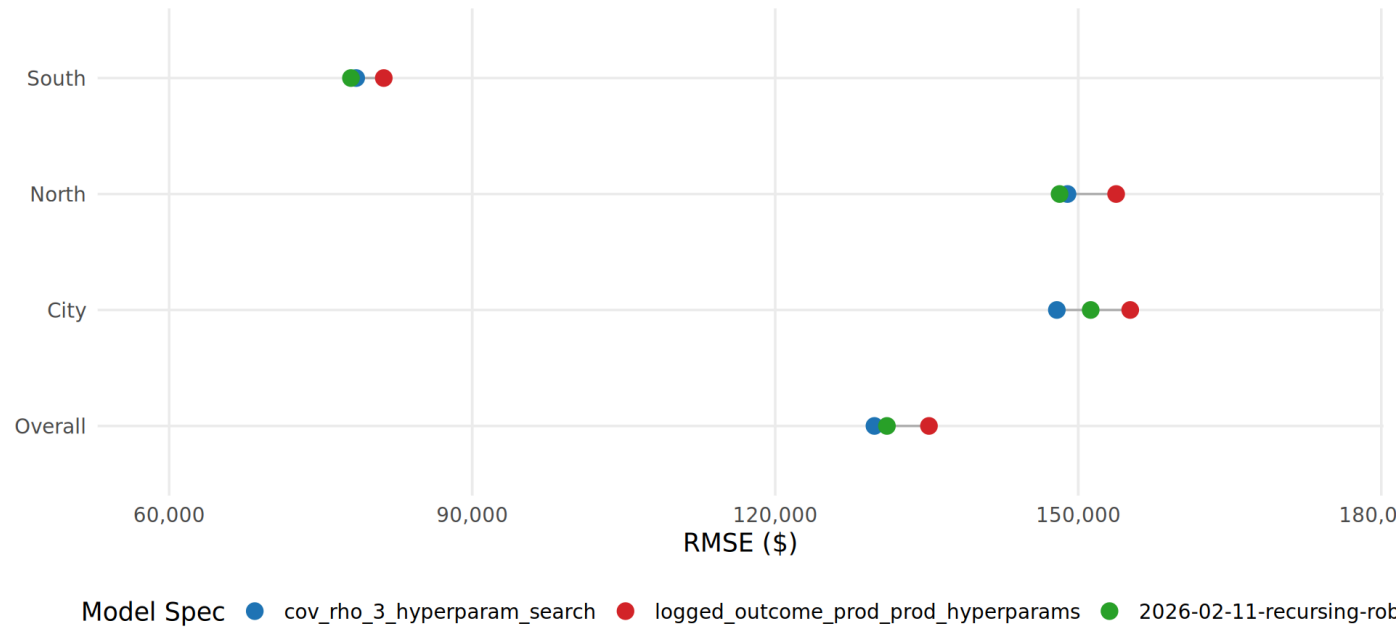
Each panel compares the three specs on a single metric — overall and in each triad.

R²



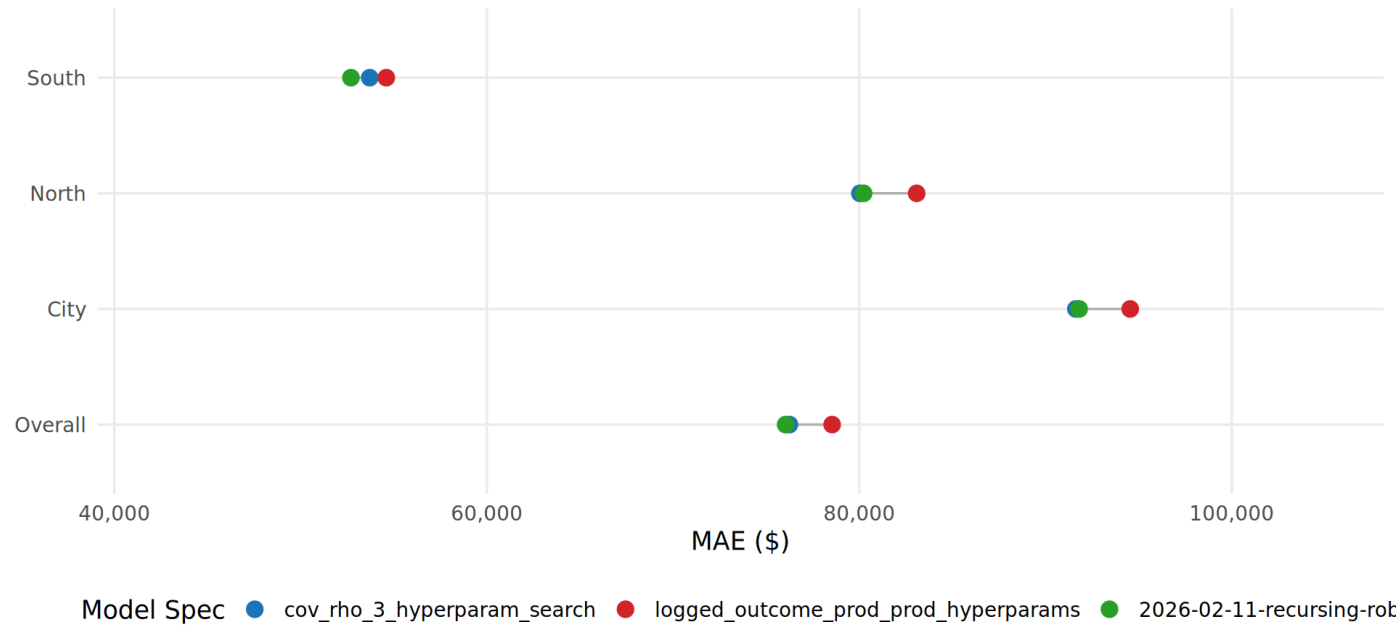
RMSE

RMSE (\$) by group and model spec

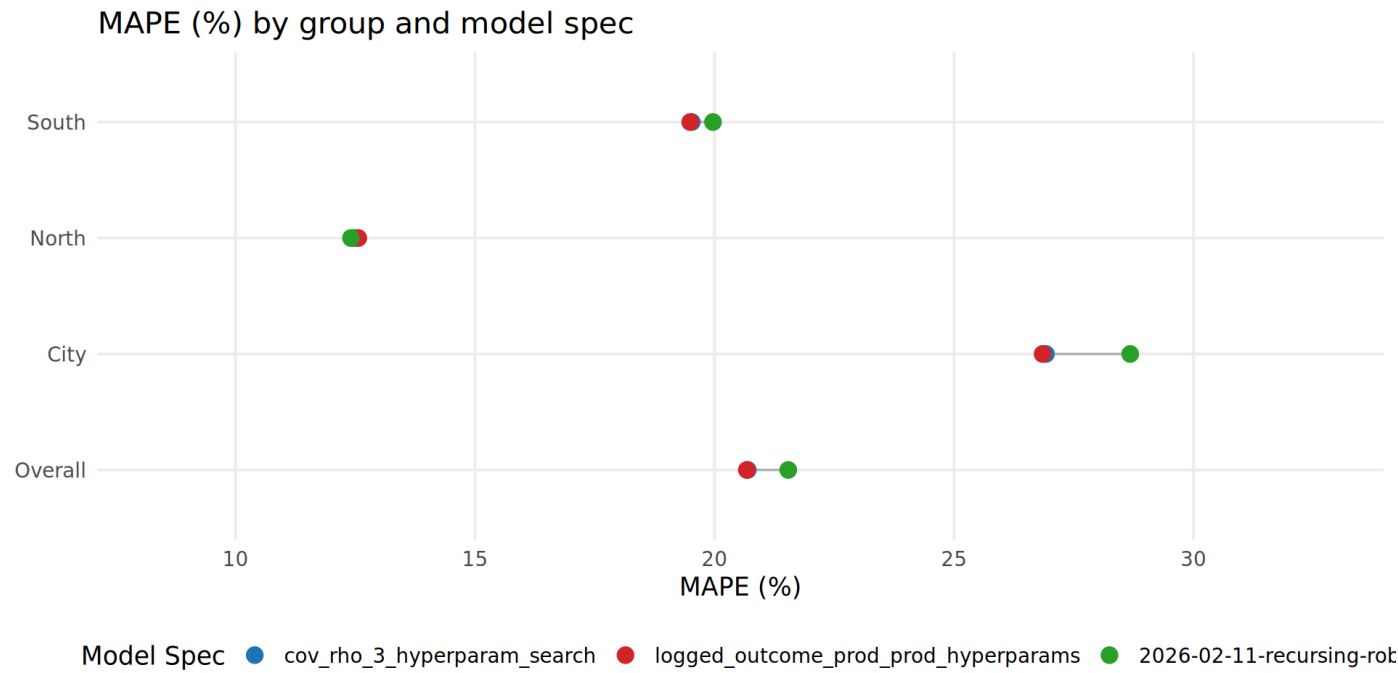


MAE

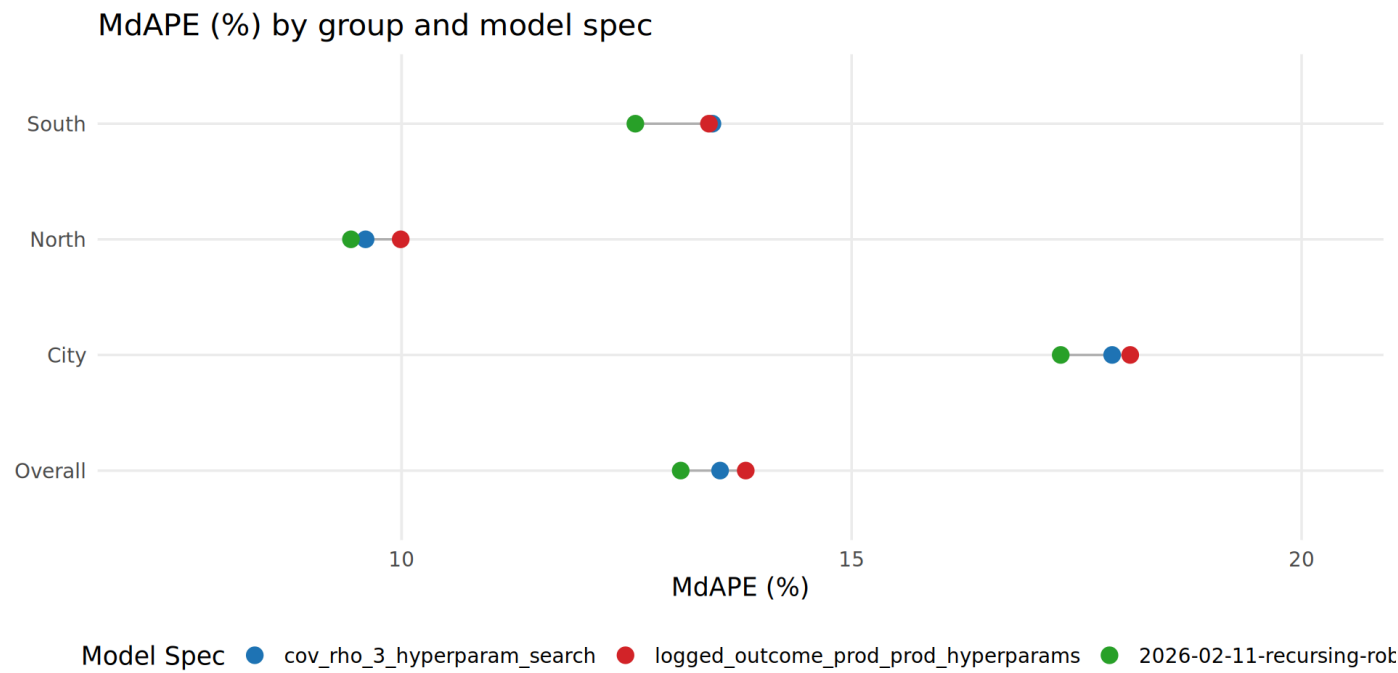
MAE (\$) by group and model spec



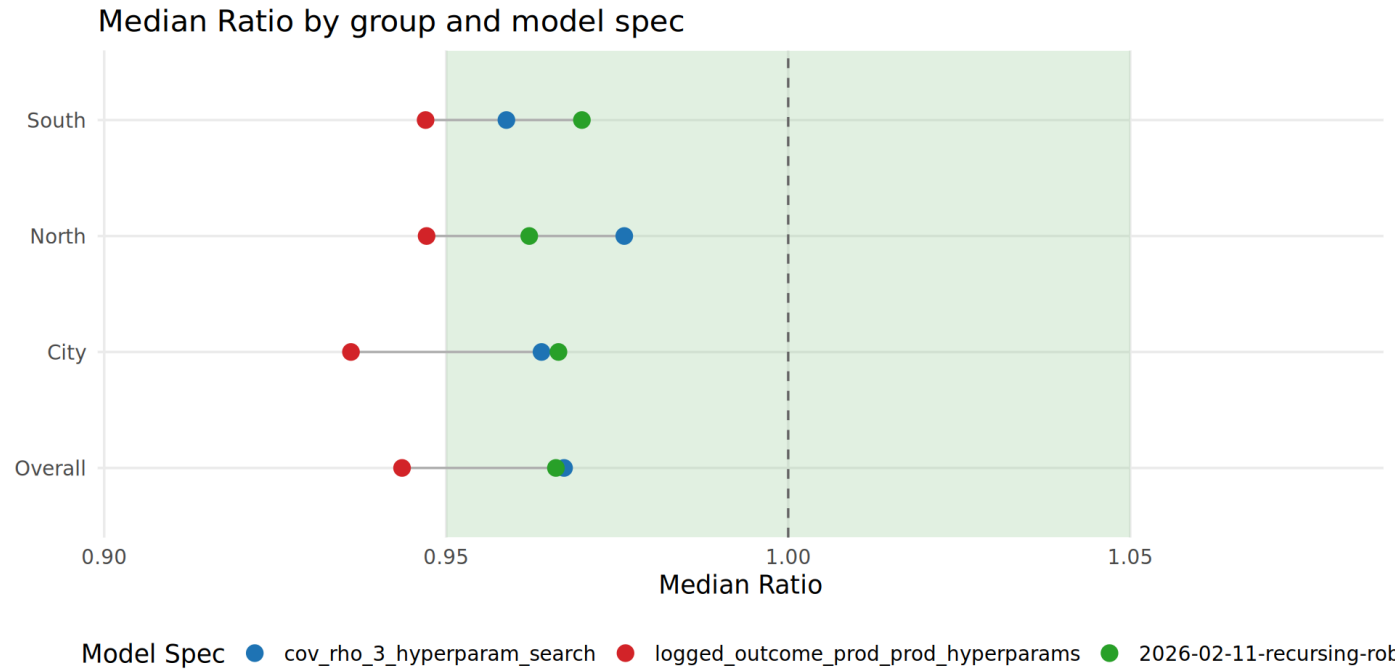
MAPE



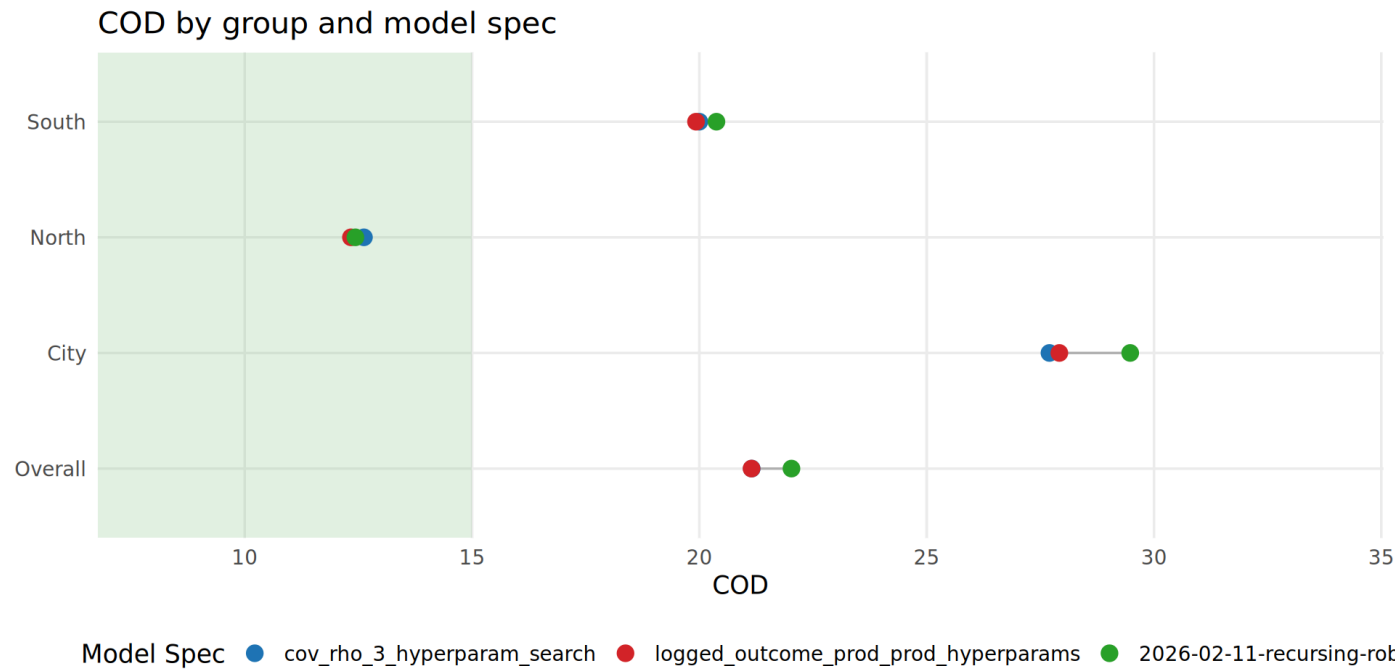
MdAPE



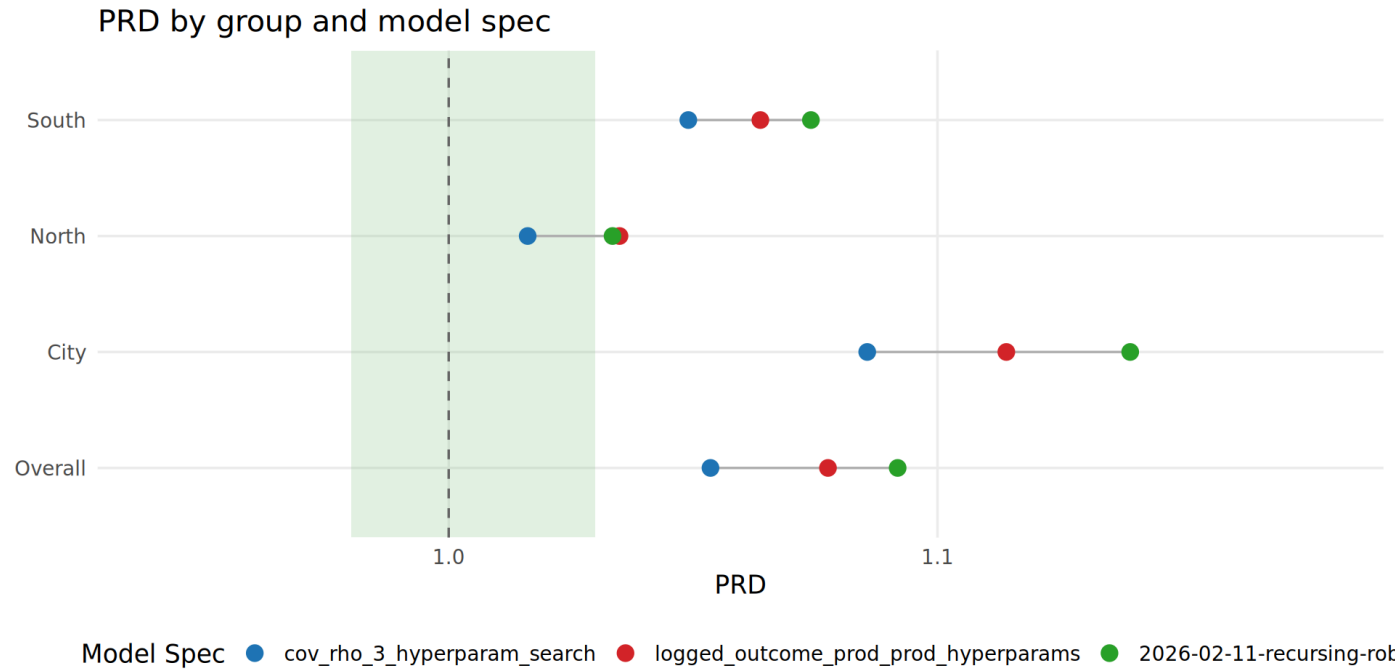
Median Ratio



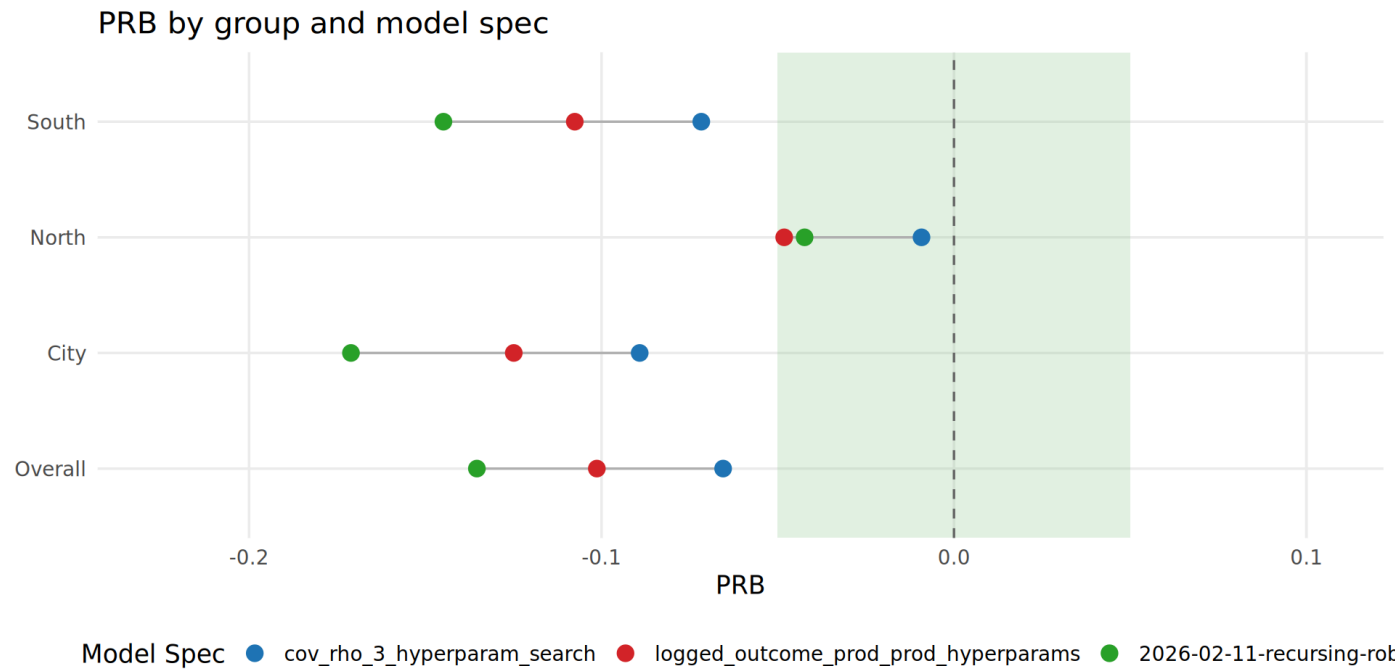
COD



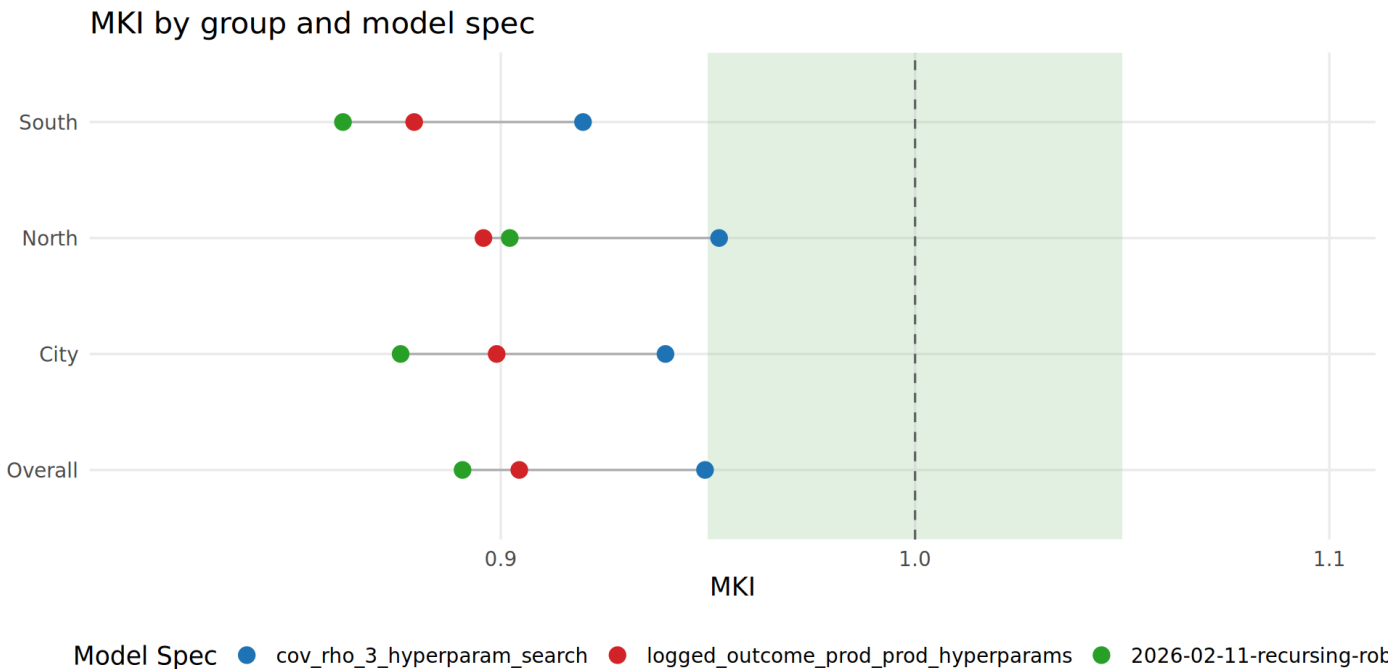
PRD



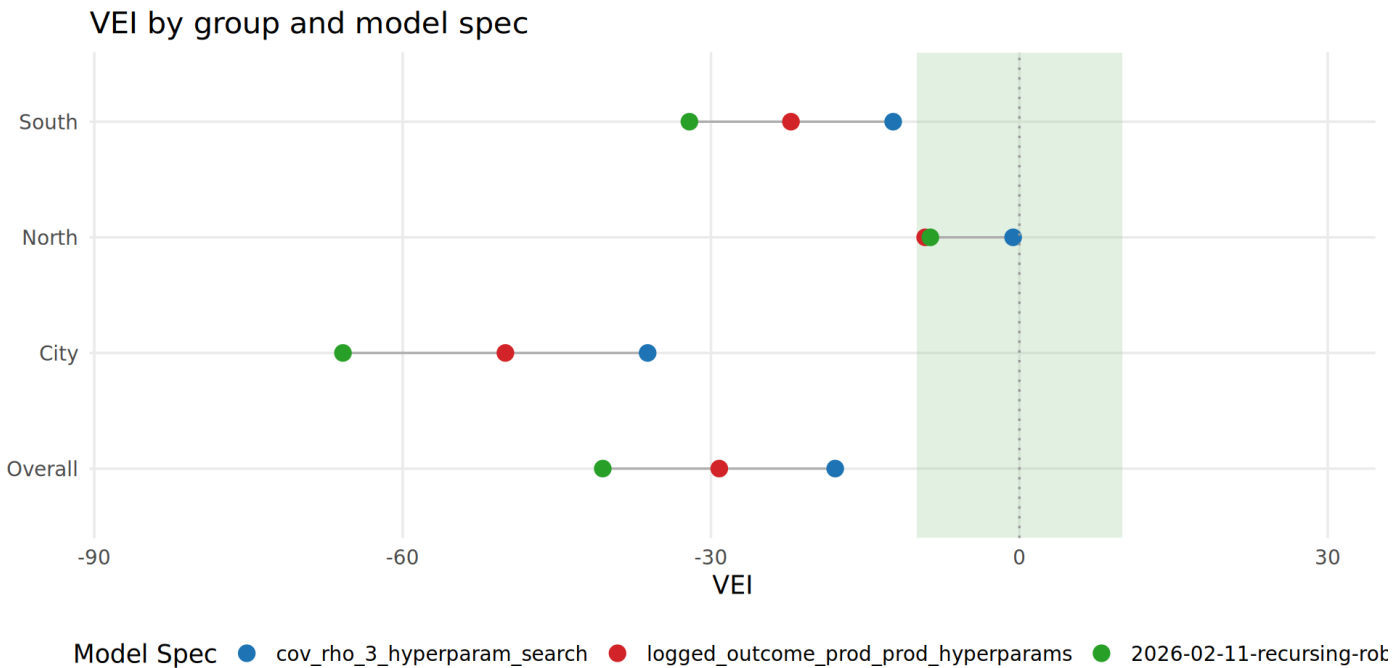
PRB



MKI



VEI



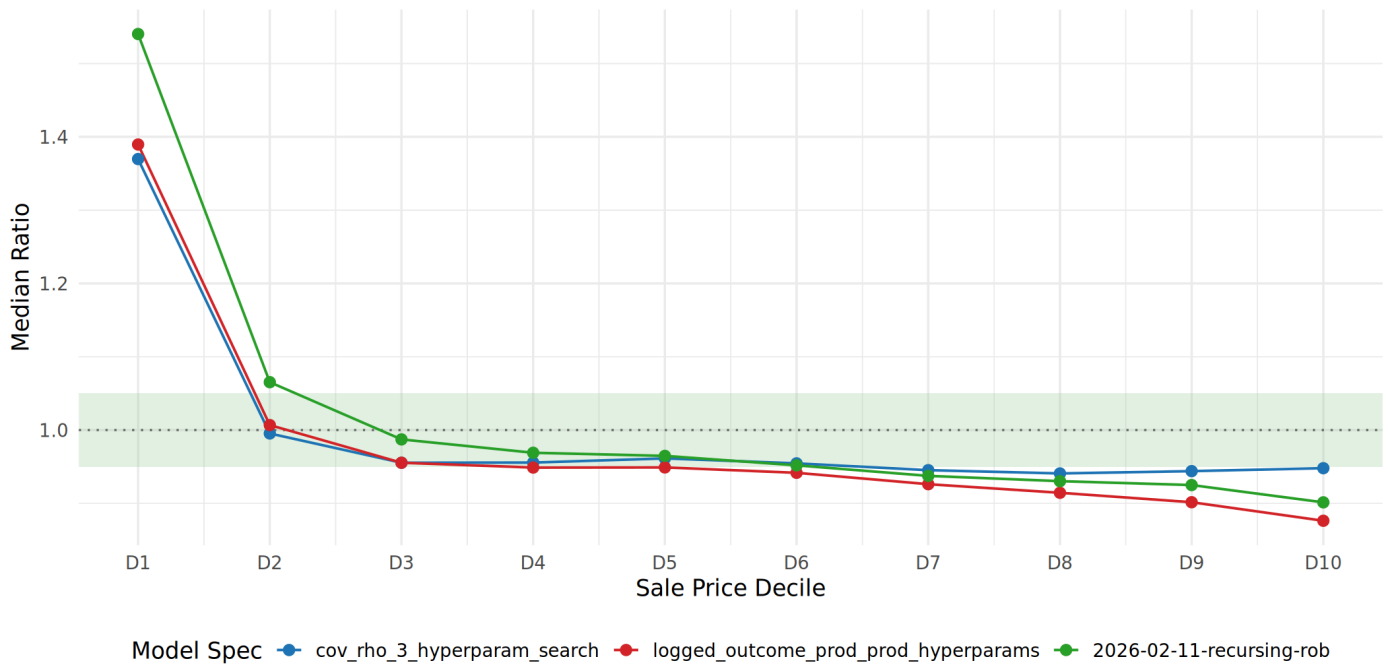
Median Ratio Decile Curves

Deciles are computed within each (model spec × group) by taking `ntile(meta_sale_price, 10)`. The green band is the IAAO target ratio band (0.95–1.05); the dotted gray line is parity (1.0).

Overall

Median Ratio by Sale Price Decile — Overall

Green band = IAAO 0.95-1.05 target. Dotted line = parity.

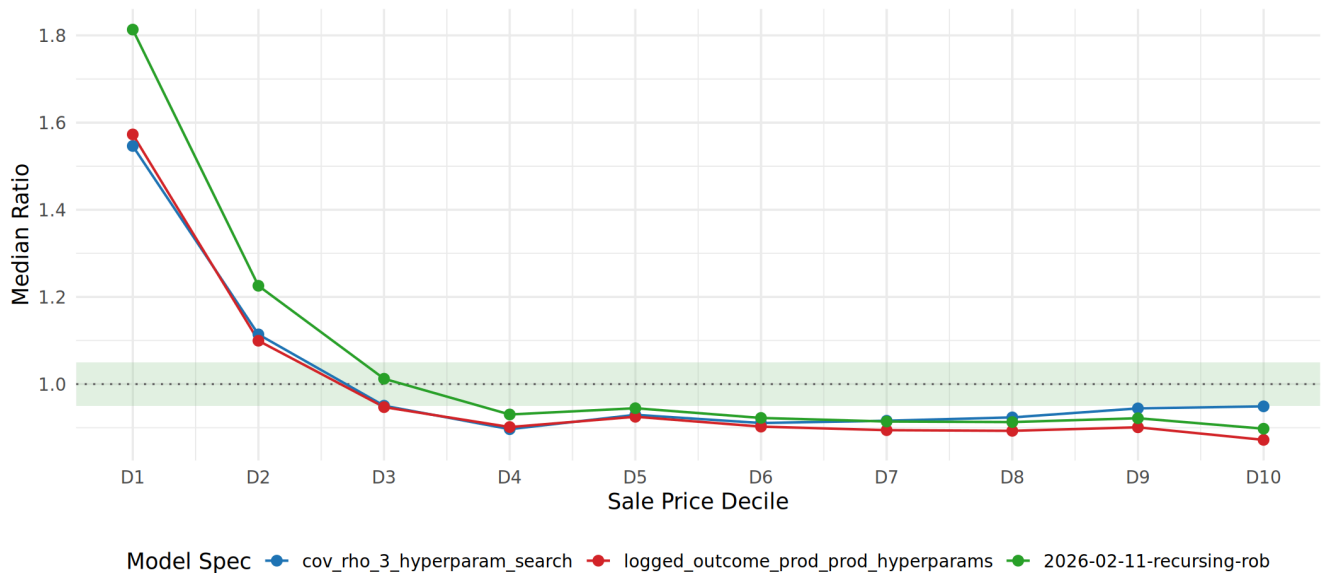


Per-Triad Individual Plots

City North South

Median Ratio by Decile — City

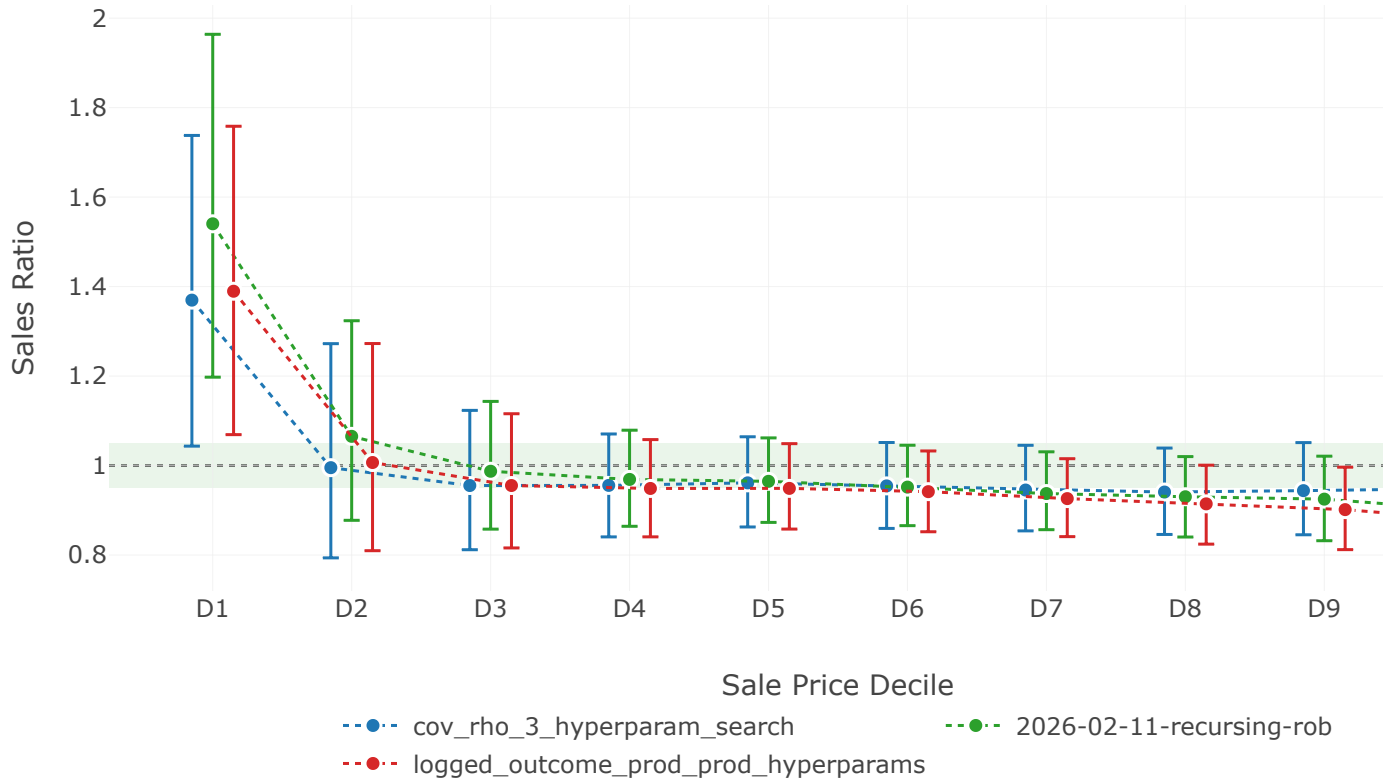
Green band = IAAO 0.95-1.05 target.



Decile Ratio Curves with Quartiles

Overall (Interactive)

Decile Ratio (Median $\pm$ IQR) — Overall



Per-Triad Individual Plots

City North South

Decile Ratio (Median $\pm$ IQR) — South

